

Finsim AI

AI-Powered Investment Learning and Market Simulator

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1. What This Project Is

Finsim AI is a web platform where users **practice investing with virtual money** while an AI explains real market behavior in plain English. It is explicitly an **education and simulation tool** — not a real trading system, not investment advice, and not a SEBI-regulated advisory product.

One-line description:

'An AI that watches real stock market data, teaches concepts using real examples, and lets users practice buying/selling with virtual money — so they understand investing before ever risking real money.'

2. Why This Name and Framing

Old Framing	Current Framing
'AI broker agent' that recommends real trades	' Simulator ' for learning
Legal gray zone (advice-adjacent, SEBI RIA concerns)	Clearly educational — no licensing required
Real trading focus	Practice + understanding focus

Positioning as a **learning simulator** (not a trading bot) keeps this legally clean and makes it a strong academic project as well as a genuine product idea. The word 'SIMULATION' or 'Virtual' stays visible directly in the UI on every relevant screen — not buried in documentation — so the legal framing holds up in practice, not just on paper.

3. Core User Flow

1. **User signs up** → gets ■1,00,000 / \$100,000 virtual cash.
2. **Platform pulls real market data** (yfinance) — never synthetic or randomly generated prices, so patterns learned actually reflect real market behavior.
3. **AI analyzes each stock** and explains signals in plain English, grounded in that stock's actual computed numbers (never generic templates).
4. **User practices buying/selling with virtual money**, with real brokerage, STT, and capital-gains tax deducted on every trade — so the numbers feel real.
5. **AI teaches the 'why'** behind each concept as it comes up, optionally read aloud via voice narration.
6. **User sees a report of virtual profit/loss**, mistakes, and lessons learned over time, alongside volatility/drawdown — not just a smoothed return figure.

4. Feature List

4.1 Core Features

- **Virtual portfolio:** Starting balance (■1,00,000 / \$100,000), buy/sell paper simulation with live P&L.;
- **Real market data:** yfinance with 2-second streaming — includes real market-regime periods (2020 crash, 2021 rally, 2022 dip, 2023-2025 bull run, 2026 live) so users experience genuine volatility.
- **Up-to-Date Financial Overview:** 69 months of institutional closing data (2021–2025) with dynamic real-time live streaming for 2026.
- **Technical indicator engine:** Wilder's 14-period RSI, EMA-12/26 MACD, 20-period 2-std dev Bollinger Bands, Volume ratio.
- **AI-generated plain English explanations:** Grounded in that stock's real computed numbers.
- **Daily virtual portfolio report:** Mini 'EOD summary' showing volatility/drawdown alongside returns.

4.2 Education-Focused Features

- **Concept lessons tied to real signals:** Concept lessons explain technical indicators the moment a stock triggers oversold or golden crosses.
- **Voice narration:** Browser's free built-in SpeechSynthesis API (no API key, zero cost).
- **Historical win-rate shown per signal type:** E.g. 'RSI < 30 has preceded a bounce ~60% of the time in backtests' — keeps every AI explanation honest.
- **Realistic costs simulation:** Brokerage + STT deducted on every virtual trade; short-term capital gains tax shown in reports.
- **Diversification score/warning:** Alerts user if virtual portfolio is over-concentrated in one stock or sector.
- **Market regime replay mode:** Real historical periods, not smoothed averages.

4.3 Engagement / Behavior Features

- **Loss 'streak' mechanic:** Consecutive losses trigger a short mandatory cooldown before the next trade, giving virtual losses behavioral weight.
- **Mandatory reflection prompt:** Required reflection after a losing trade before the next action is allowed.
- **Progress tracking:** Virtual P&L; history, lessons completed, mistakes flagged over time.

4.4 Data & Scale Approach

- **Starter watchlist:** ~20 liquid, well-known NSE stocks across sectors for the initial build.
- **Scaled up to full market coverage:** 2,400+ securities (all 2,298+ active Indian NSE stocks in  plus 100+ Global US market leaders in \$).
- **AI explanations cached:** Cached per stock per day and served to all users watching that stock, staying comfortably within free-tier LLM limits.

4.5 Implemented & Enhanced Architecture

- **Full NSE market screener & catalog:** 2,400+ securities with instant search, sector tabs, and 2-second streaming ticks.
- **Multi-user authentication:** Mandatory Full Name validation on sign-up before OTP or Google verification.
- **Interactive Alerts & Signals Drawer:** Real unread count tracking, category badges (AI SIGNAL, TRADE FILL, SYSTEM), and filter tabs.
- **Complete Dark / Light Mode:** Dynamic chart strokes and Tailwind CSS v4 custom variant.
- **Optional real-broker paper trading connection:** Zerodha Kite Personal API protocol (free tier) for advanced users, strictly human-approved only.

5. Tech Stack

Layer	Technology	Cost
Frontend Hub	Next.js 16.3.4 (React 19), Tailwind CSS v4, Lucide React, Recharts	Free / Open Source
Backend / Tech Engine	Python 3.10+, Flask, Pandas, NumPy	Free / Open Source
Database	SQLite3 (investiq.db, finsim.db) → PostgreSQL (production)	Free
Stock Data	yfinance (sub-2s streaming in-memory cache)	Free
Full NSE Stock List	NSE official equity list CSV (2,298+ equities) + Global US catalog	Free
AI Explanations	Groq LLM (Llama 3.1) / OpenAI-compatible API	Free tier
Voice Narration	Browser SpeechSynthesis API	Free, built-in
Auth	Flask-Login & Next.js session cookies (with mandatory Full Name)	Free
Hosting	Render / Railway (backend), Vercel (frontend)	Free tier
Scheduler	GitHub Actions / Background daemons	Free

Total cost to build and run at small scale: 

6. Project File Structure

```
Invest_iq/
■■■■ app.py # Flask app & API reverse-proxy gateway
■■■■ data_provider.py # yfinance data fetching, live/mock toggle
■■■■ scanner.py # Runs technical analysis across watchlist/market
■■■■ indicators.py # RSI, MACD, Bollinger, volume ratio calculations
■■■■ explainer.py # Builds plain-English explanations from real numbers
■■■■ signal_stats.py # Tracks and returns historical win-rate per signal type
■■■■ portfolio.py # Virtual portfolio logic – buy/sell, P&L, fees, tax
■■■■ auth.py # User accounts and sessions
■■■■ models.py # Database schema (users, trades, watchlists, signal history)
■■■■ broker_kite.py # Zerodha Kite Connect API paper bridge
■■■■ build_history.py # 2021-2026 multi-year benchmark history extractor
■■■■ all_monthly_history.json# 69-month institutional closing data (Nifty 50 & S&P; 500)
■■■■ generate_pdf.py # Automated Project Overview PDF generator
■■■■ Invest_IQ_Project_Overview.pdf # Generated PDF Project Overview
■
■■■■ frontend/ # Next.js 16 (React 19) Full-Stack Web Application
■ ■■■■ src/app/ # App Router (Dashboard, Markets, Trade, Learn, Auth)
■ ■■■■ src/components/ # Topbar (Bell+Theme), Sidebar, NotificationsDrawer, PortfolioChart
■ ■■■■ src/lib/ # financial-history.ts, market-api.ts, store.tsx, catalogs
■
■■■■ templates/ # Flask Jinja2 templates (dashboard, screener, portfolio, report)
■■■■ static/ # CSS & static assets for Flask engine
■■■■ tests/ # Automated verification test suite
■■■■ requirements.txt # Python package dependencies
```

7. Build Roadmap & Status

Phase	Goal	Status
1	Data pipeline — fetch real historical stock data via yfinance with retry handling	■ Completed
2	Technical indicators — RSI, MACD, Bollinger Bands, volume ratio computed and stored	■ Completed
3	Signal detection + plain English explainer grounded in real numbers with visible disclaimer	■ Completed
4	Virtual portfolio — buy/sell simulation with real brokerage, STT, and capital-gains tax deducted per trade	■ Completed
5	Historical win-rate tracking per signal type, surfaced transparently next to every AI explanation	■ Completed
6	Behavioral guardrails — loss-streak cooldown and mandatory reflection prompt after a losing trade	■ Completed
7	Voice narration (browser SpeechSynthesis API) on explanations and reports	■ Completed
8	Authentication — real user accounts with mandatory Full Name validation on sign-up	■ Completed
9	Scale stock list to full market — 2,400+ securities (2,298+ NSE + 100+ Global mega-caps) with 2s live streaming	■ Completed
10	Up-to-date Financial Overview — 69 months real institutional close data (2021–2025) and dynamic live 2026 streaming	■ Completed
11	Interactive alerts & dark/light mode — slide-over drawer with unread tracking and Tailwind v4 theme sync	■ Completed
12	Cloud deployment configuration (Docker, Render, Vercel, Railway) ready for staging & production	■ Completed

8. Legal Positioning

- Finsim AI is always framed as:
- A **learning and simulation tool**, not investment advice.
 - Virtual money only in the core product — any future real-money connection remains fully human-approved, with no autonomous trading.
 - Every AI-generated signal explanation carries a clear disclaimer and its real historical win-rate — never a confident directive to 'buy' or 'sell'.
 - The word '**SIMULATION**' or '**Virtual**' visible directly in the UI on every relevant screen.

This framing keeps the project outside SEBI Registered Investment Adviser (RIA) requirements, which apply to personalized real-money investment advice, not education/simulation tools.